3.4.1 The equation of a circle when the centre and radius are given

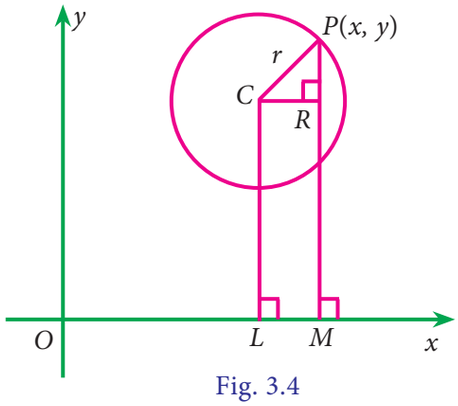


Fig. 3.4

Let $C(h, k)$ be the centre and 'r' be the radius of the circle

Let P(x, y) be any point on the circle

$$CP = r$$

$$CP^2 = r^2$$

$$(x-h)^2 + (y-k)^2 = r^2$$

is the equation of the circle.

In particular, if the centre is at the origin, the equation of circle is $x^2 + y^2 = r^2$

Example 3.17

Find the equation of the circle with centre at (3, –1) and radius is 4 units.

Solution

Equation of circle is

$$(x-h)^2+(y-k)^2 = r^2$$

Here $(h, k) = (3, -1)$ and $r = 4$

Equation of circle is

$$(x-3)^2+(y+1)^2 = 16$$

$$x^2-6x+9+y^2+2y+1 = 16$$

$$x^2+y^2-6x+2y-6 = 0$$

Example 3.18

Find the equation of the circle with centre at origin and radius is 3 units.

Solution

Equation of circle is $x^2 + y^2 = r^2$

Here $r = 3$

i.e equation of circle is $x^2 + y^2 = 9$

3.4.2 Equation of a circle when the end points of a diameter are given

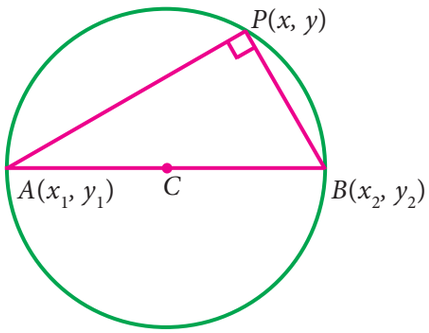


Fig. 3.5

Let $A(x_1, y_1)$ and $B(x_2, y_2)$ be the end points of a diameter of a circle and

$P(x, y)$ be any point on the circle.

We know that angle in the semi circle is 90°

$$\therefore \angle APB = 90^\circ$$

$\therefore$ (Slope of AP) (Slope of BP) = –1

$$\left(\frac{y-y_1}{x-x_1}\right)\times\left(\frac{y-y_2}{x-x_2}\right) = -1$$

$$(y-y_1)(y-y_2) = -(x-x_1)(x-x_2)$$

$\Rightarrow (x-x_1)(x-x_2)+(y-y_1)(y-y_2) = 0$ is the required equation of circle.

Example 3.19

Find the equation of the circle when the end points of the diameter are (2, 4) and (3, –2).

Solution

Equation of a circle when the end points of the diameter are given is

$$(x-x_1)(x-x_2)+(y-y_1)(y-y_2) = 0$$

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